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Research note

Tuning the parameters of an artificial neural network using central composite design and genetic algorithm

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Artificial neural networks;
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Desirability function.

Abstract An artificial Neural Network (ANN) is an efficient approach applied to solving a variety of problems. The main problem in using ANN is parameter tuning, because there is no definite and explicit method to select optimal parameters for the ANN parameters. In this study, three artificial neural network performance measuring criteria and also three important factors which affect the selected criteria have been studied. Moreover, central composite design has been used to design experiments and also analyze network behavior according to identified parameters, by using the overall desirability function. Then the Genetic Algorithm has been proposed to find optimal parameter status. For this purpose, the proposed method has been illustrated by the numerical example of a well known mathematical function. The results show that the designed ANN, according to the proposed procedure, has a better performance than other networks by random selected parameters and also parameters which are selected by the Taguchi method. In general, the proposed approach can be used for tuning neural network parameters in solving other problems.

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1. Introduction

Artificial neural networks are used for solving a variety of problems, such as finance, manufacturing, electronic, medical etc. [1]. In general, the backpropagation method is used for training neural network [2,3]. Gradient descent, conjugate gradient descent, resilient, BFGS quasi-Newton, one-step secant, Levenberg–Marquardt and Bayesian regularization are some different algorithms that can be used for training in the backpropagation method [4]. During the design and training of an ANN, some parameters must be defined, like the number of hidden layers and the number of neurons in each layer; these features change upon the specific application. There is no general and explicit method for choosing these parameters. Almost a trial and error method is used, but it needs more computational time and is not a precise method. Hence, proposing an approach is necessary to find the optimum combination of parameters to affect the performance of ANN.

The second section briefly reviews some approaches to artificial neural network tuning. In Section 3, the proposed method is given. In Section 4 the proposed approach is applied to a mathematical numerical example. Finally, conclusions are made in Section 5.

2. Literature review

Many authors have used different approaches in tuning the parameters of heuristic and meta heuristic algorithms. Shi and Eberhart [5] have determined guidelines for selecting parameters of a particle swarm optimizer, and validate guidelines by using the design of experiments. Xu et al. [6] have applied statistical tests to tune the Tabu search algorithm. Bashiri and Karimi [7] applied a general factorial design for tuning the Tabu search for solving the Quadratic Assignment Problem.

Several authors have used the design of experiments in order to find an optimum combination of parameters, which would have an effect on the performance of ANN. Khaw et al. [8] applied the Taguchi methodology by using two sets of simulated

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data to determine ANN's effective parameters to increase the accuracy and convergence speed of BPN. Peterson et al. [9] used the Taguchi methodology to investigate causes of error in BPN. Yang and Lee [10] applied the Taguchi methodology to minimize the time of training in ANN. Packianather et al. [11] used the Taguchi methodology to show the effect of network design parameters and neural network behavior for a wood veneer inspection. Yum and Kim [12] applied the Taguchi's dynamic methodology with consideration of noise conditions to design multilayer feedforward neural networks. Sukthomya and Tannock [13] applied the Taguchi methodology to identify the optimum combination of effective parameters in ANN, and demonstrated the use of this approach in manufacturing. Tortum et al. [14] used the Taguchi methodology to investigate the optimum combination of effective parameters in ANN performance, and also demonstrated the effect of each parameter in performance criteria.

All mentioned approaches are based on the Taguchi methodology for tuning the parameters of ANN and finding the optimum combination of effective parameters in the performance of ANN.

Myers et al. [15] demonstrated that the response surface methodology is a proper alternative to the Taguchi method. It is clear that in the Taguchi methodology, the optimum value of parameters is situated in one of the determined levels of the parameter. Hence, using other alternative methods seems to be necessary to find more precise values of the optimized ANN design parameters.

Another technique to design structure and learning parameters of ANN is using Evolutionary Algorithms (EA). EA can be used to assist in ANN design and training. However, the lack of information and tools for analysis of the results, which is achieved by these techniques make the work in this area difficult [16].

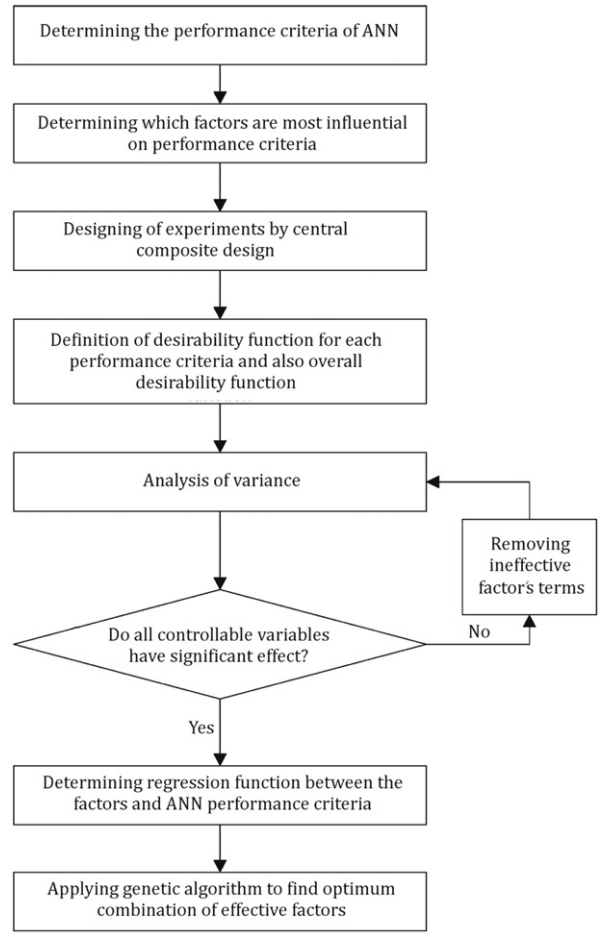

Figure 1: Flow chart of proposed approach.

3. Proposed approach

3.1. Determining the performance criteria of ANN

This section contains the proposed methodology which has been illustrated in Figure 1.

In the first step, we have to determine some criteria which measure the performance of ANN correctly. ANN performance criteria considered in this study are shown in Table 1. Their definition has been illustrated in the table, as well.

The first considered performance criterion is Root Mean Square Error (RMSE) between targets and outputs of the neural network. It is clear that this response is "the Smaller The Better (STB)" type.

The second criterion is the correlation coefficient (R-value) between the outputs and targets in ANN. It is a measure of how well the variation in the outputs is explained by the targets. This variable takes a value between 0 and 1. If this number is close to 1, then there is a good correlation between targets and outputs, and if it is close to zero, then there is a poor correlation. Hence, this response can be considered "the Larger The Better (LTB)" type.

The ANN training time spent between the beginning and end of the training process is the other important criterion to measure the ANN performance. It is clear that, like RMSE, this response is "the Smaller The Better" type.

3.2. Determining which factors are most influential on performance criteria

Our goal in this study is finding the optimum combination of controllable factors which would affect the performance of ANN. Hence, it is crucial to determine controllable factors that would have the most important effects on the performance of ANN. These factors are described as follows:

(A) The percentage of training data: The data are divided into three subsets: training, validation and testing set. The training set is used for computing the gradient and updating the network weights and biases. As the network begins to over-fit the data, the error on the validation set typically begins to rise. After the specified number of iterations (which can be defined by the researcher) and the validation error increases, the training is stopped, and the weights and biases of the epoch, with minimum validation errors are returned as the final ANN structure.

In this study, the percentage of validation data is considered 5% of the data which is assumed to be a constant value. In other words, for example, if the parameter level is taken as 0.7, it means that 70% of the data are used for training the network, while 25% and 5% of the data are used for testing and validating, respectively.

(B) The number of neurons in the first layer: This factor is one of the most effective parameters in performance of ANN. Although more neurons require more computation, their

Table 1: Performance criteria of ANN.

RMSE	Root of mean square error between the outputs and targets
R-value	Correlation coefficient between the outputs and targets in ANN
TT	Time between the beginning and the end of training ANN

implementation might result more efficiently for solving complex problems.

(C) The number of neurons in the second layer: This factor determines the number of neurons in the second layer, and moreover, helps us to determine whether the network is one layered or not. In other words, for example, if the parameter level is taken as zero, it means that the network has only one hidden layer. Although more layers require more computation, their implementation might result more efficiently for solving complex problems.

3.3. Designing of experiments by central composite design

Central Composite Design (CCD), also called response surface methodology, is a rapid technique which extracts the relationship between responses and controllable factors [17]. To fit a complete second-order model for controllable factors, CCD is a logical method for experimental design [18]. The Central Composite Design includes a 2^k factorial, $2k$ axial or star and n_c center runs. For designing experiments by CCD, two parameters should be specified. These two parameters are α and n_c. For example, for 2 controllable factors, the points in the axial (star) portion of the design are at $(+\alpha, 0)$, $(-\alpha, 0)$, $(0, +\alpha)$ and $(0, -\alpha)$. In this study, because of finding the controllable factors as integer values, parameter α is defined as 2.

3.4. Definition of desirability function for each performance criterion and also the overall desirability function

To optimize several responses simultaneously, a desirability function technique is applied [19]. The desirability function transforms the value of response to scale free-value, and denotes it as d_i for the ith response. The desirability function value is between 0 and 1. The closer d_i is to one, the more desirable the response is [20]. Derringer and Suich [21] defined this function for a Nominal-The Best (NTB) type response as Eq. (1).

$$d_i(\hat{y}_i(x)) = \begin{cases} 0 & \text{if } \hat{y}_i(x) < Y_i^{\min} \\ & \text{or } \hat{y}_i(x) > Y_i^{\max} \\ \left[\dfrac{\hat{y}_i - Y_i^{\min}}{T_i^{\min} - Y_i^{\min}}\right]^{s_i} & \text{if } Y_i^{\min} \le \hat{y}_i(x) \le T_i^{\min} \\ \left[\dfrac{Y_i^{\max} - \hat{y}_i}{Y_i^{\max} - T_i^{\max}}\right]^{t_i} & \text{if } T_i^{\max} \le \hat{y}_i(x) \le Y_i^{\max} \\ 1 & T_i^{\min} \le \hat{y}_i(x) \le T_i^{\max} \end{cases} \tag{1}$$

where $d_i(\hat{y}_i(x))$ is the desirability function of the ith response and $Y_i^{\min}$ and $Y_i^{\max}$ are the lower and upper bounds of the response, respectively. $T_i^{\min}$ and $T_i^{\max}(T_i^{\min} \le T_i^{\max})$ are the lower and upper desirable values of the ith response, respectively, and s_i and t_i are the parameters which determine the shape of the desirability function: if s_i (or t_i) $= 1$, the shape is linear; if s_i (or t_i) > 1, concave; and if $0 < s_i$ (or t_i) < 1, convex. It should be mentioned that if $T_i^{\min} = T_i^{\max}$ the shape of $d_i(\hat{y}_i(x))$ transforms from trapezoidal to triangular.

Harrington [22] proposed a geometric mean in order to aggregate the individual desirability functions, and approach to overall desirability, function, D. Then, the optimal combination set of factors is determined by maximizing D. In this study, the weighted geometric mean, which is proposed by Derringer [23], is shown in Eq. (2):

$$D = \left(d_1^{W_1} d_2^{W_2} \cdots d_I^{W_I}\right)^{\frac{1}{\sum W_i}}, \tag{2}$$

where w_i is the relative weight of the ith response.

3.5. Analysis of variance and determining the regression function between factors and ANN performance criteria

For analysis of variance, the overall desirability function is defined as a single response. ANOVA determines which of the considered controllable factors are more effective. Factors which are recognized as ineffective are removed, and ANOVA is reused for the remaining effective factors. By implementing the ANOVA, regression coefficients for overall desirability and effective factors are extracted.

3.6. Applying genetic algorithm to find optimum combination of effective factors

The Genetic algorithm has been proved to be a successful method for solving LP and NLP problems, inspired by the process of natural selection and genetic evaluation. GA applies mutation, crossover and selecting operators to a population of encoded parameter space. The algorithm explores different areas of the parameter space, and directs the search to the region where a high probability of global optimum [24,25] exists. To study more about genetic algorithms see [26,27]. In the proposed method, after establishing the regression relation between the overall desirability function and effective factors, the genetic algorithm is applied for finding the optimum combination set of effective factors.

4. A numerical example

In this example, the proposed method is used for tuning the artificial neural network parameters, for training the peaks function. For this approach, 400 data which are distributed uniformly between -3 and 3, are used. In general, Levenberg–Marquardt backpropagation algorithm convergence properties are better than others [28]. Thus, a Levenberg–Marquardt backpropagation algorithm is used for all experiments in this study.

The peaks function is defined in Eq. (3), and its surface is shown in Figure 2.

$$f(x, y) = 3(1-x)^2 e^{(-x^2-(y+1)^2)} - 10\left(\frac{x}{5} - x^3 - y^5\right) e^{(-x^2-y^2)} - \frac{1}{3} e^{(-(x+1)^2-y^2)}. \tag{3}$$

4.1. Determining the performance criteria of ANN

Performance criteria were described in Section 3. These criteria were defined in Table 1.

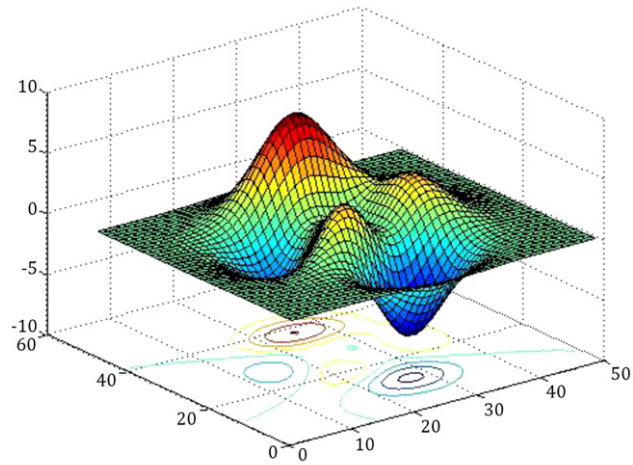

Figure 2: Peaks function.

Table 2: Parameters and their levels studied in the experiments.

Factors		Cube points		Central points	Axial points	
		Low	High		Low	High
A	The percentage of trained data	0.675	0.825	0.75	0.6	0.9
B	The number of neuron in first layer	17	49	33	1	65
C	The number of neuron in second layer	15	45	30	0	60

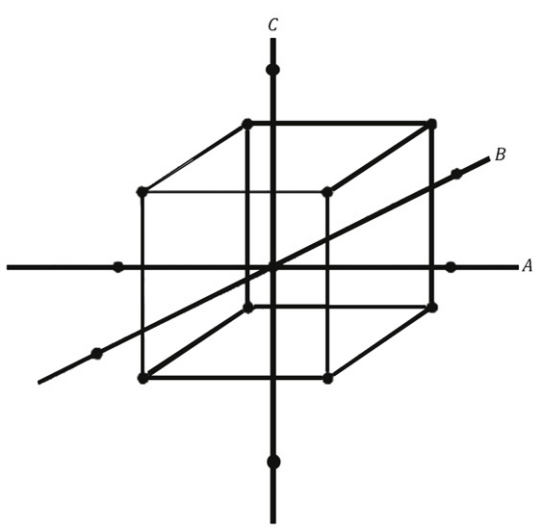

Figure 3: Central composite design for three factors.

4.2. Determining which factors are most influential on performance criteria

Three factors which are mostly influential on ANN performance were described in the proposed method section, but determining their levels is crucial for designing the experiments. The axial, cube and central point in a central composite design for three factors is shown in Figure 3. In this example, the factor's lower and upper bounds are specific.

Hence, designing an experiment by considering axial points is logical, and central points and cube points are defined by considering axial points and α parameter. Factors with their own axial, central and cube points are shown in Table 2.

Table 3: Results of central composite design (CCD).

Base runs	20	Cube points	80
Replicates	10	Center points in cube	60
Total runs	200	Axial points	60
Alpha	2	Center points in axial	0

4.3. Designing of experiments by central composite design

After defining levels of effective factors, the designing of experiments by central composite design can be done. In this example, the design of experiments is created by MINITAB statistical software. Results of CCD for this example are shown in Table 3. As seen, 10 replications for each experiment are experiments by CCD are shown in Table 4. All results of response variables for considered. Designed each treatment of designed experiments were computed by MATLAB software, according to the designed neural network. The experimental results for three responses are shown separately in Appendix.

4.4. Definition of the desirability function for each performance criterion and also overall desirability function

By considering the desirability function in Eq. (1) and also the type of responses in this study, the desirability function for RMSE, R-value and training time is formulated as follows:
Desirability function for RMSE

$$d_i(\hat{y}_i(x)) = \begin{cases} 0 & \text{if } \hat{y}_i(x) > Y_i^{\max} \\ \left[\dfrac{Y_i^{\max} - \hat{y}_i}{Y_i^{\max} - T_i}\right]^{t_i} & \text{if } T_i \le \hat{y}_i(x) \le Y_i^{\max}. \end{cases} \tag{4}$$

Desirability function for R-value

$$d_i(\hat{y}_i(x)) = \begin{cases} 0 & \text{if } \hat{y}_i(x) < Y_i^{\min} \\ \left[\dfrac{\hat{y}_i - Y_i^{\min}}{T_i - Y_i^{\min}}\right]^{s_i} & \text{if } Y_i^{\min} \le \hat{y}_i(x) \le T_i. \end{cases} \tag{5}$$

Table 4: Experimental plans designed by CCD.

Run	Parameters and their levels		
	A	B	C
1	0.750	33	30
2	0.900	33	30
3	0.825	49	45
4	0.750	33	30
5	0.750	33	60
6	0.675	17	45
7	0.675	49	45
8	0.750	33	30
9	0.750	33	30
10	0.675	49	15
11	0.675	17	15
12	0.750	33	0
13	0.600	33	30
14	0.750	33	30
15	0.825	17	15
16	0.750	33	30
17	0.750	1	30
18	0.825	17	45
19	0.750	65	30
20	0.825	49	15

Table 5: Lower and upper bounds of responses and targets.

Responses	Lower and upper bounds of responses and targets
RMSE	$T_{\max} = T_{\min} = Y_{\min} = 0, Y_{\max} = 0.1$
Total R-value	$T_{\min} = T_{\max} = Y_{\max} = 1, Y_{\min} = 0.7$
Train time	$T_{\min} = Y_{\min} = 0, T_{\max} = 10, Y_{\max} = 60$

Table 6: Weights of responses.

Responses	RMSE	R-value	Train time
Weights	2	2	1

Desirability function for training time

$$d_i(\hat{y}_i(x)) = \begin{cases} 0 & \text{if } \hat{y}_i(x) < Y_i^{\min} \text{ or } \hat{y}_i(x) > Y_i^{\max} \\ \left[\dfrac{Y_i^{\max} - \hat{y}_i}{Y_i^{\max} - T_i^{\max}}\right]^{t_i} & \text{if } T_i^{\max} \le \hat{y}_i(x) \le Y_i^{\max} \\ 1 & \text{if } T_i^{\min} \le \hat{y}_i(x) \le T_i^{\max}. \end{cases} \tag{6}$$

By considering desirability functions, their shapes can be plotted in Figure 4. Desirability function parameters for selected responses are shown in Table 5. For example, for the first response (RSME), if y_i takes a value more than 0.1, then its desirability value should be zero.

For calculating the overall desirability function, weights of each response should be determined. In this numerical example, selected weights are shown in Table 6. In this example we have supposed that training time is not as important as the two others. Anyway, response weights can be defined by a decision maker.

4.5. Analysis of variance and determining the regression function between factors and ANN performance criteria

After creating the individual desirability function and the overall desirability function D, we have to find a relation function between D and each controllable factor. The result of analysis of variance by Minitab statistical software is considered for establishing the regression function. Estimated regression coefficients extracted from the analysis of variance for overall desirability function D, for all terms, are shown in Table 7.

It is extracted from Table 7 that the percentage of training data (factor A) is not an effective factor in performance of ANN. Hence, we can omit the terms in which factor A is involved, and perform ANOVA for remaining terms again. Estimated regression coefficients extracted from the analysis of variance for overall desirability function D, for remaining terms, are shown in Table 8, and the analysis of variance is illustrated in Table 9.

Table 7: Estimated regression coefficients for D considering all of the terms.

Term	Coef.	SE coef.	T	P
Constant	−0.41532	2.07731	−0.200	0.842
A	−1.65896	5.02869	−0.330	0.742
B	0.06221	0.02119	2.936	0.004
C	0.04493	0.02259	1.988	0.48
A^*A	1.22438	3.24279	0.378	0.706
B^*B	−0.00041	0.00007	−5.791	0.000
C^*C	−0.00040	0.00008	−4.896	0.000
A^*B	−0.01964	0.02695	−0.729	0.467
A^*C	−0.00110	0.02874	−0.038	0.970
B^*C	−0.00027	0.00013	−2.028	0.044
$S = 0.2892$,	$R - \text{Sq} = 49.5\%$,	$R - \text{Sq(adj)} = 47.1\%$		

Table 8: Estimated regression coefficients for D considering effective terms.

Term	Coef.	SE coef.	T	P
Constant	−0.975139	0.175236	−5.565	0.000
B	0.047861	0.006274	7.629	0.000
C	0.044497	0.006673	6.668	0.000
B^*B	−0.000418	0.000069	−6.020	0.000
C^*C	−0.000403	0.000079	−5.103	0.000
B^*C	−0.000273	0.000135	−2.032	0.0044
$S = 0.2888$,	$R - \text{Sq} = 48.6\%$,	$R - \text{Sq(adj)} = 47.3\%$		

Table 9: Analysis of variance for D.

Source	DF	Seq. SS	Adj. SS	Adj. MS	F	P
Regression	5	15.2910	15.2910	3.05819	36.67	0.000
Linear	2	10.5185	5.6887	2.84435	34.11	0.000
Square	2	4.4282	4.4282	2.21411	26.55	0.000
Interaction	1	0.3442	0.3442	0.34423	4.13	0.044
Residual error	194	16.1791	16.1791	0.08340		
Lack-of-fit	3	0.1729	0.1729	0.05764	0.69	0.561
Pure error	191	16.0062	16.0062	0.08380		
Total	199	31.4701				

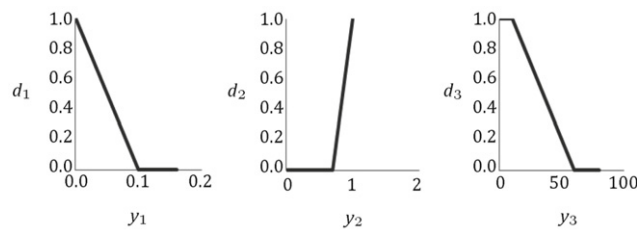

Figure 4: Shapes of desirability functions.

Table 10: Initial parameters of genetic algorithm.

Population size	Initialize population	Scaling function	Selection function	Crossover operator	Percent of cross over	Mutation operator	Percent of mutation	Stopping criteria
100	Randomly	Rank	Tournament	Two point cross over	$P_c = 0.75$	Gaussian mutation	$P_m = 0.01$	Iteration = 1000

Table 11: Optimum values of factors B and C, using the proposed method.

The number of neurons in first layer	The number of neurons in second layer	Overall desirability function
44	40	0.9761

By considering Table 8, the regression function between overall desirability function D and factors B and C is formulated as Eq. (7).

$$\hat{D} = -0.975139 + 0.047861B + 0.044497C - 0.000418B^2 - 0.000403C^2 - 0.000273BC. \quad (7)$$

The effects of influential parameters are illustrated in Figure 5. As seen, by increasing the number of neurons in both first and second layers of ANN, the overall desirability function is increased up to a specific value and then decreased.

4.6. Applying the genetic algorithm to find the optimum combination of effective factors

In this example, a genetic algorithm by toolbox of MATLAB software is applied for maximizing the overall desirability function and finding optimum values of factors. Initial parameters for solving a problem by genetic algorithm are illustrated in Table 10. Results of optimization by genetic algorithm are shown in Table 11.

In order to confirm the results of the proposed method, ANN is trained in considered situation with five replications. As mentioned before, percentage of training data is not recognized as an effective factor in performance of ANN. Hence, in these five replications for confirmation of the results, percentage of training data is considered in its central point value of 0.75 (see Table 2). The results of these confirmation experiments are shown in Table 12.

For validating the goodness of results which are gained by the proposed method, a comparison is done between the proposed method and the Taguchi approach. So, a $L_{16}(2^3)$ orthogonal array of experimental design is chosen. Final results of optimization of effective factors, using the Taguchi method for this example, are illustrated in Table 13. Experiments (ANN training and results) are conducted five times under optimum

Table 12: Results of confirmation experiments for the proposed method.

Responses	1st repeat	2nd repeat	3rd repeat	4th repeat	5th repeat	Mean	d_i	D
RMSE	9.22e−05	5.17e−15	6.59e−14	0.0003222	4.582e−09	8.29e−05	0.9992	0.9983
R-value	0.99928	0.99861	0.99910	0.99946	0.99828	0.99895	0.9965	
Train time	10	7	8	8	6	8.25	1	

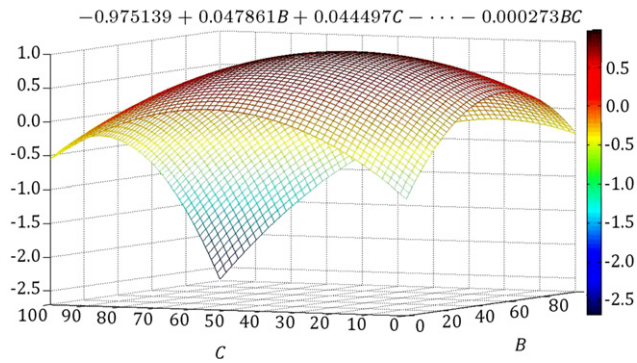

Figure 5: Overall desirability response surface.

Table 13: Optimum values of factors B and C, using the Taguchi method.

The percentage of trained data	The number of neurons in first layer	The number of neurons in second layer
0.675	49	45

Table 14: Results of confirmation experiments for the Taguchi method.

Responses	1st repeat	2nd repeat	3rd repeat	4th repeat	5th repeat	Mean	d_i	D
RMSE	0.00044	6.55e−15	0.00011	0.06732	9.1e−06	0.01357	0.8642	0.9345
R-value	0.99643	0.99699	0.997138	0.99760	0.99603	0.99684	0.9895	
Train time	10	11	11	11	12	11.27	0.9745	

conditions, and results of these experiments are illustrated in Table 14. A comparison between the proposed approach result and the Taguchi design, based on Tables 12 and 14, shows that the proposed approach outperforms previous approaches.

5. Conclusion

Parameter selection is a main problem when using artificial neural networks. Previous works which have been already proposed often used the Taguchi method. But, the main disadvantage of previous studies is their discrete solution space. For compensating this flaw, a central composite design is proposed. CCD can be applied for tuning the parameters in continuous solution space. In this study, for validating the proposed method, an example with results of performing the mentioned approach has been presented. Results of the example demonstrate that ANN has a good performance under optimum conditions, which are determined by the proposed method. For demonstrating the goodness of the results of the proposed method, a comparison between the Taguchi

Table A.1: Experimental results for RMSE using the proposed method.

Run	1st repeat	2nd repeat	3rd repeat	4th repeat	5th repeat	6th repeat	7th repeat	8th repeat	9th repeat	10th repeat
1	0	0.0264	0	0	0.0222	0.0292	0.0121	0.0132	0.0011	0.0023
2	0.0045	0.017	0.0021	0.0008	0.0179	0	0.0391	0.0054	0.0003	0.0014
3	0.0075	0	0.0319	0.0001	0	0.022	0.0019	0.0007	0	0.0054
4	0.0215	0	0.0003	0	0.0005	0.0023	0	0.0035	0	0.0346
5	0.0175	0.0319	0	0.0176	0	0.0003	0	0	0.0007	0.0085
6	0	0.0004	0.0042	0.0293	0	0.0002	0.0001	0.0006	0.0066	0.1581
7	0	0	0.0006	0.0101	0	0.0031	0.0396	0	0	0.0027
8	0	0.056	0	0	0.0009	0.001	0.0809	0.0035	0.0063	0.0925
9	0.0006	0.0255	0.0168	0.0005	0	0.0064	0.0671	0.0002	0	0
10	0.0586	0.0475	0.033	0.0052	0.0086	0.1679	0.0042	0.007	0.0011	0.005
11	0.1562	0.2227	0.0953	0.0735	0.009	0.8911	0.0298	0.4254	0.0065	0.2538
12	0.183	0.065	0.1162	0.0587	0.0515	0.0511	0.2313	0.0437	0.2007	0.2933
13	0	0.0002	0	0.0011	0.0012	0	0.0048	0.0016	0.0006	0.003
14	0.4195	0.0007	0.0002	0.0011	0.098	0	0.0001	0.0004	0.0037	0.0005
15	0.0206	0.1353	1.3892	0.0154	1.7916	0.1992	0.2462	0.0028	0.0002	0.0005
16	0	0	0.0001	0	0.0505	0.0063	0.0848	0.0002	0	0.0195
17	1.9183	1.6793	1.5067	1.241	1.5906	1.6093	1.8412	1.4765	1.9261	1.49
18	0	0.0003	0.1269	0.0001	0.0028	0.0003	0.0145	0.0023	0.0053	0.0035
19	0.0123	0	0	0.153	0.019	0.0086	0	0.0002	0.0051	0.0072
20	0.0065	0.008	0.0318	0.0018	0.0292	0	0.0019	0	0.2888	0.0151

Table A.2: Experimental results for total *R*-value using the proposed method.

Run	1st repeat	2nd repeat	3rd repeat	4th repeat	5th repeat	6th repeat	7th repeat	8th repeat	9th repeat	10th repeat
1	0.99936	0.99967	0.99965	0.99953	0.99988	0.99949	0.99773	0.99961	0.99783	0.99951
2	0.99965	0.99995	0.99996	0.99989	0.9999	0.99998	0.99989	0.99996	0.99997	0.99998
3	0.99905	0.99903	0.9993	0.99931	0.99878	0.99963	0.99835	0.99946	0.9998	0.99953
4	0.99977	0.99975	0.9997	0.99956	0.99978	0.99957	0.9998	0.99959	0.99971	0.99956
5	0.99901	0.99856	0.99938	0.99858	0.99892	0.9991	0.99846	0.99899	0.99895	0.99906
6	0.99964	0.9999	0.99781	0.99965	0.99971	0.99914	0.99976	0.99995	0.99975	0.99091
7	0.99683	0.99789	0.99797	0.99593	0.99792	0.99689	0.99719	0.99865	0.99617	0.9978
8	0.99991	0.99961	0.99979	0.99963	0.99943	0.99878	0.99853	0.99961	0.99981	0.99766
9	0.99935	0.99944	0.99967	0.9997	0.99848	0.99981	0.99888	0.99969	0.99973	0.99991
10	0.99448	0.99886	0.99743	0.99829	0.99945	0.99622	0.99897	0.99849	0.99902	0.99956
11	0.99337	0.96351	0.99678	0.99676	0.99972	0.88733	0.99886	0.95597	0.99982	0.99042
12	0.99424	0.99848	0.99782	0.99809	0.99918	0.99902	0.99377	0.99943	0.99386	0.99333
13	0.99776	0.99476	0.9977 0	0.99842	0.99893	0.99871	0.99911	0.99557	0.99545	0.99947
14	0.98829	0.99565	0.99893	0.99965	0.99866	0.99951	0.99959	0.99958	0.99979	0.99988
15	0.99986	0.99612	0.69464	0.99983	0.38343	0.99824	0.98433	0.99999	0.99999	1
16	0.99975	0.99981	0.99941	0.99912	0.9982	0.99951	0.99905	0.99932	0.99942	0.99917
17	0.55529	0.43019	0.56397	0.71092	0.62358	0.64301	0.59543	0.62271	0.47877	0.43992
18	0.99996	0.99996	0.99847	0.99997	0.99937	0.99999	0.99995	0.99995	0.99972	0.99999
19	0.99942	0.99812	0.99859	0.99804	0.99906	0.99833	0.99886	0.99911	0.99882	0.99856
20	0.99974	0.99968	0.99923	0.9999	0.99894	0.99959	0.99982	0.99965	0.99675	0.9992

Table A.3: Experimental results for total train time using the proposed method.

Run	1st repeat	2nd repeat	3rd repeat	4th repeat	5th repeat	6th repeat	7th repeat	8th repeat	9th repeat	10th repeat
1	18	17	17	19	20	17	17	17	14	42
2	41	18	17	109	19	20	12	16	18	18
3	21	16	21	19	16	16	22	14	174	18
4	8	12	10	13	72	11	12	14	11	18
5	58	25	15	17	21	25	19	21	34	15
6	14	19	16	18	17	25	49	122	35	16
7	25	15	13	15	15	15	15	18	15	12
8	18	21	13	9	13	9	19	15	19	26
9	32	13	12	12	79	13	19	13	40	10
10	20	12	14	12	11	12	10	18	12	13
11	22	18	13	21	18	15	18	19	61	13
12	15	20	37	41	104	81	11	73	22	11
13	9	14	48	9	24	10	13	25	18	14
14	17	27	10	18	16	15	20	22	11	12
15	19	20	18	24	8	9	20	46	220	404
16	16	15	38	30	15	12	12	16	9	16
17	6	6	8	7	7	7	6	9	6	27
18	696	18	16	14	99	20	25	25	37	16
19	13	13	11	13	22	11	13	11	11	11
20	13	12	16	13	18	170	19	61	14	14

approach and the proposed method is performed. Results of the comparison show that all the considered performance criteria have a better situation in the proposed approach, rather than in the Taguchi method. As a future study, other experimental designs can be used for better ANN parameter tuning. Moreover, other parameters can be studied for the tuning of artificial neural networks.

Appendix

In this section, experimental data for the illustrated example in the text have been reported in Tables A.1–A.3.

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